

Module 5 & 6 Group Discussion

ECE 525.642 FPGA Design using VHDL

VGA Research, Test, and Verification

The group discussion for this week will apply to Lab 4 to help gain more insight into VGA displays and how they function. Lab 4 has a major component that needs to be designed, which is to display a grid pattern on a VGA display. For this to work, the Hsync and Vsync need to be setup properly with a valid window for drawing the checkboard pattern. Since the VGA component is large and large components should be simulated and tested prior to integration with the rest of the system, it's a good idea to add a testbench for this component.

For this group discussion, please work together as a group to research the VGA specification, work out the proper signaling specification for the signals, and a test plan for verifying the sync signals in your design. Some tasks for the group can be:

- Research the 640x480 VGA specification, discuss the history, the refresh rate, the proper sync signal timing. Note: Some of this is in the lab assignment, so make sure to expand on this and provide a clear understanding of the VGA interface. Provide some history and references to support your research.
- Discuss the overall test plan for the VGA controller and what kinds of inputs and outputs would be needed for a proper test. Write the testbench skeleton with the module instantiated and placeholders for the verification process.
- Design a process or entity to verify the output of a VGA controller is valid. Spend time on validating the sync signals (Multiple students can work together on this aspect)

Grading for this discussion will be on participation and value of the participation within your group as well as the quality of the end output. The testbench does not have to fully function but work towards that goal is important. The goal is to provide thoughtful discussion within the group to help everyone work on validating their Lab 4 VGA designs.

Note: Make sure not to work on the VGA controller itself, as that is an individual effort for each student and should NOT be discussed in the group setting. Make sure to keep the discussion to validating the VGA component from an external point of view only! Only discuss the input and output signals to the component and validating the output.

